Zijian Cao

ROBOTICS &
MECHANICAL DESIGN

MSc Robotics student | The University of Manchester

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PROFILE

MSc Robotics student with a mechanical engineering background. Experience includes CAD concepts, team robot prototyping and programming, and numerical aerofoil analysis using XFOIL and ANSYS Fluent.

EDUCATION

MSc Robotics Sep 2026 - Aug 2027 (expected)

The University of Manchester

Mechanical Design, Manufacturing and Automation Sep 2022 - Jun 2026

Joint undergraduate programme: North China University of Technology and Brunel University of London.

PROJECT EXPERIENCE

Comparative aerodynamic study of NACA aerofoils Report: Apr 2026

Undergraduate final project

- Compared NACA 2412 and NACA 4412 using XFOIL and selected two-dimensional ANSYS Fluent simulations.
- Analysed lift, drag, pitching moment and flow-field outputs to discuss aerofoil-selection trade-offs for small fixed-wing UAV applications.
- Documented model limitations, including two-dimensional scope and the absence of experimental validation.

Internal pipe climber prototype Spring 2025

ME2628 team design-and-build project

- Contributed to mechanical design, assembly and programming within a five-person team.
- The team built a physical prototype and documented motor, mechanism and connection troubleshooting, infrared-sensor setup and vertical-pipe trials in a development video.

Ball-throwing vehicle Feb - May 2024

Team course project

- Mainly responsible for programming, alongside design and assembly contributions within the team.
- Developed a gear-driven arm concept for ball pickup and throwing, and produced a multi-view CAD assembly drawing using the Makeblock parts library.

SKILLS IN USE

Programming work: main programming responsibility in the ball-throwing vehicle team project.
Simulation: XFOIL and ANSYS Fluent, used in the final project.
Practical engineering: CAD assemblies and drawings; component selection and BOM cost planning; vehicle design and assembly.